

or proxies (Hubinger et al., 2019; Ivanitskiy et al., 2023) such as internal reward models (Marks et al., 2023b). While current state-of-the-art AI systems as prediction-trained LLMs are considered relatively safe (Hubinger et al., 2023), progress on interpreting RL systems may prove critical for safeguarding the next paradigm (Aschenbrenner, 2024).

Acknowledgements

I am grateful for the invaluable feedback and comments from Leon Lang, Tim Bakker, Jannik Brinkmann, Can Rager, Louis van Harten, Jacqueline Bereska, Benjamin Shaffrey, Thijmen Nijdam, Alice Rigg, Arthur Conny, and Tom Lieberum. Their insights substantially improved this work.

Glossary

- circuits** Sub-graphs within neural networks consisting of *features* and the weights connecting them. Circuits can be thought of as *computational primitives* that perform understandable operations to produce (ideally interpretable) features from prior (ideally interpretable) features. Examples include circuits for detecting curves at specific orientations (Cammara et al., 2020; 2021), continuing repeated patterns in text (Olsson et al., 2022), and resolving anaphoric references (Wang et al., 2023). While circuits can involve clearly interpretable features, the definition allows for intermediate representations that are less easily interpretable.. 3, 9, 10, 20, 26, 27, 35
- concepts** An abstract idea or representation derived from observations of the world. Concepts refer to the *natural abstractions* that a cognitive system, like a neural network, aims to capture and represent through its learned *features*, which may or may not align perfectly with human-defined concepts.. 4, 7, 8, 32, 34
- deceptive alignment** When a misaligned model aims to appear aligned to gain more power to take control once sufficiently powerful.. 24
- deceptive inflation** Theoretical result on deceptive behavior: policies produce trajectories that look better than they actually are from the human’s perspective with limited observations to get higher reward signals during training. This deceptive behavior arises in reinforcement learning from human feedback when the human provides feedback based only on partial observations of the trajectories, while the policy has full state information during training (Lang et al., 2024).. 28
- disentangled** In disentangled representations, individual dimensions or components correspond to distinct, *independent factors of variation in the data*, rather than representing a tangled mixture of these factors.. 4, 9, 15, 16, 20, 27, 32, 33
- eliciting latent knowledge** Developing strategies to make a machine learning model explicitly report latent facts or knowledge embedded in its parameters, especially in cases where the model’s output is untrusted (Christiano et al., 2021). This involves finding patterns in neural network activations that track the true state of the world (Mallen & Belrose, 2023).. 24
- features** The fundamental units of how neural networks encode knowledge, which cannot be further decomposed into smaller, distinct *concepts*. Features are core components of a neural network’s representation, analogous to how cells form the fundamental unit of biological organisms (Olah et al., 2020). The *superposition* hypothesis suggests an alternative definition: that features correspond to the *disentangled* concepts that a larger, sparser network with *sufficient capacity* would learn to represent with individual (*monosemantic*) neurons (Olah et al., 2020; Bricken et al., 2023).. 3, 4, 7, 8, 10, 13, 16, 18, 20, 23, 27, 28, 32, 33, 35
- grokking** "Grokking refers to the surprising phenomenon of delayed generalization where neural networks, on certain learning problems, generalize long after overfitting their training set." (Liu et al., 2022a). 21, 22
- hydra effect** The phenomenon where models can internally self-repair and maintain capabilities even when key components are ablated, making it challenging to identify the relevant components underlying a particular behavior (McGrath et al., 2023).. 18, 26
- inner misalignment** Inner misalignment, or goal misgeneralization, occurs when an AI system develops goals or behaviors during training that are misaligned with the intended objectives despite a correctly specified reward signal.. 24, 28
- internal world models** Internal causal environment models formed within neural networks, implicitly emerging as a by-product of prediction (e.g., in large language models).. 11, 12, 24, 34

irreducible We adopt the notion of *features* as the fundamental units of neural network representations, such that features cannot be further decomposed into smaller, distinct factors. To make this more precise, we can formalize the definition of features as irreducible input patterns following Engels et al. (2024): A feature f of sparsity s is a function that maps a subset of the input space (with probability $1 - s > 0$) into a higher-dimensional representational space. We say the feature is active on this subset. A feature f is reducible into features a and b if there exists a transformation that decomposes f into a and b , such that the transformed distribution $p(a, b)$ is either:

1. Separable: $p(a, b) = p(a)p(b)$
2. A mixture: $p(a, b) = wp_1(a, b) + (1 - w)p_2(a, b)$ where p_1 is lower-dimensional.

Features are defined as irreducible patterns that cannot be decomposed into separable or mixture distributions via such transformations. This formalizes the notion that features form the fundamental atomic units underlying neural representations. Features that can be *disentangled* into statistically independent components (separable) or simpler lower-dimensional factors (mixtures) are not considered the core representational primitives. The key properties are that 1) features map from the input space to higher-dimensional representational spaces, 2) features are sparse and only activated on subsets of the input, and crucially, 3) features are irreducible and cannot be expressed as transformations of other statistically independent components.. 4

iterative distillation and amplification A technique for training AI systems by repeatedly distilling knowledge from a larger model into a smaller one while amplifying the smaller model’s capabilities through feedback and interaction with humans.. 24

linear representation Features are directions in activation space, i.e., linear combinations of neurons.. 3, 8, 14, 28

machine unlearning Techniques for removing private data or dangerous knowledge from models.. 25

mesa-optimization The emergence of unintended subagents within a model with their own objectives, potentially misaligned with the original training objective.. 24

microscope AI Systems that extract and utilize knowledge from a model without allowing the model to take autonomous actions. This involves reverse engineering a trained model to understand its learned knowledge about the world, aiming to leverage this understanding directly without deploying the model in an operational capacity.. 24

modularity The property of an AI system being composed of distinct, semi-independent components or submodules that can be separately understood, modified, and recombined, rather than a monolithic, opaque structure.. 20, 21

monosemantic A neuron corresponding to a single concept. The intuition is that analyzing what inputs activate a given neuron reveals its associated semantic meaning or concept. In contrast to *polysemantic*.. 5, 6, 14, 16, 20, 32, 34

motifs Repeating patterns that emerge across models and tasks, manifesting as circuits, features, or higher-level behaviors from component interactions. Examples include curve detectors, induction circuits, and branch specialization. Motifs reveal common structures and mechanisms underlying neural network intelligence.. 3, 10, 11, 21, 35

natural abstractions High-level summaries or descriptions of a system or environment learned and used by many cognitive systems. According to the *natural abstraction hypothesis* (Chan et al., 2023), a set of "natural" abstractions exist that represent redundantly encoded information in the world and tend to be learned by intelligent systems produced through local selection pressures. These natural abstractions form a relatively small, discrete set of concepts like "tree," "velocity," etc., that allow compact descriptions of the world while discarding many irrelevant low-level details.. 4, 10, 32